Jordan Cao

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Experience

Mercury Jan 2026 – May 2026
Software Engineering Intern *San Francisco, CA (Remote)*

- Designed and implemented pause/unpause functionality for autotransfer rules for **300,000+ users** in a large-scale Haskell fintech backend, spanning database schema changes, domain types, SQL queries, and REST API handlers.
- Built database migrations and schema evolution for autotransfer rule lineage tracking, including backfill scripts and a new pause table with indexing, ensuring **zero-downtime deployments** on production banking infrastructure.
- Shipped the pause/unpause autotransfer feature end-to-end, exposing pause status and history through the API with full HSpec test coverage; **adopted by over 70% of customers** who use autotransfers.
- Participated in Agile sprint planning, stand-ups, and peer code reviews, and contributed to CI/CD pipelines to ship autotransfer disablement logic for partner bank account migrations touching **millions of dollars** in transfers, using Temporal workflows to guarantee reliable execution across service boundaries in a Nix-based monorepo.

Machine Learning & Evolutionary Algorithms Research Assistant Sep 2025 – January 2026
Wilfrid Laurier University, Supervised by Dr. Azam Asilian Bidgoli *Waterloo, Kitchener*

- Extracted 1024-dimensional feature vectors for 140,000+ image patches across 533 whole-slide images using Phikon-2, then applied Leiden clustering to build slide-level representations for classification.
- Designed a Genetic Algorithm to select the most impactful clusters via crossover and mutation, cutting computational cost while preserving model accuracy (full results below, under Projects).

Data Specialist Jan 2025 - April 2025
Wilfrid Laurier University *Waterloo, Kitchener*

- Built predictive models for donor segmentation, identifying high-propensity prospects with ∼**85% accuracy** to support targeted outreach.
- Created interactive dashboards and automated reporting tools for stakeholders using structured SQL queries to support real-time analytics.
- Validated and cleaned 100,000+ alumni records, removing duplicates and resolving inconsistencies, **improving reporting accuracy by 15%**.

Projects

GTOtutor | *TypeScript, React, Fastify, PostgreSQL, Rust, AWS, Claude API* Live | GitHub

- Built and deployed a full-stack poker trainer that grades beginners' decisions against GTO charts across heads-up and 6-max tables, using a **custom Rust preflop solver** and live turn/river solves on AWS Lambda.
- Integrated a Claude-powered AI coach that explains solver decisions and identifies player leaks, with Supabase Auth and Postgres storing hands, session reviews, and a player style profile.
- Hosted on AWS (S3 + CloudFront, Dockerized Fastify API on EC2) with a mobile-optimized React 19 UI covered by Playwright end-to-end tests.

Whole-Slide-Image (WSI) Classification Study | *Python, Leiden Clustering, Genetic Algorithms, Phikon-2*

- Found that Leiden clustering on raw features outperformed PCA-based K-means, achieving a **weighted F1-score of 0.944** versus ∼0.83 for the best K-means baseline.
- Designed a Genetic Algorithm for evolutionary feature selection, **reducing the feature space by 95% (201 to 10 features)** while retaining a weighted F1-score of 0.905, cutting computational cost.

Education

BSc in Computer Science Waterloo, ON
Wilfrid Laurier University *Sep 2026*

Skills

Languages: Haskell, Java, Python, C/C++, Rust, SQL, JavaScript, TypeScript, HTML/CSS, VBA, Assembly
Developer Tools/Specialized Skills: Git, Nix, Docker, PostgreSQL, React, Redux, Node.js, Fastify, REST APIs, CNNs, Image Classification, Data Processing, Supabase, AWS, Power BI, Temporal, Cursor, Claude Code, Figma
Libraries: Esqueleto, Persistent, HSpec, Pandas, NumPy, Matplotlib, PyTorch, Scikit-learn, TensorFlow